

Tracking the Evolution of the Hemoglobin Beta (HBB) Gene Across Species

Project Objective

To investigate the evolutionary conservation of the **Hemoglobin Beta (HBB)** gene across six (I used 12 speices) species using sequence retrieval, BLAST, pairwise and multiple sequence alignment, sequence logo generation, and phylogenetic analysis.

Task 1: Sequence Retrieval & BLAST Search

Reference Query:

- **Human HBB Protein**
 - **Accession Number:** NP_000509.1
 - **Organism:** *Homo sapiens*

BLAST Search:

- Tool used: **NCBI BLASTp**
- Database: **Non-redundant protein sequences (nr)**
- Target species:
 - *Pan troglodytes*, *Gorilla gorilla*, *Macaca mulatta*, *Chlorocebus sabaeus*, *Rhinopithecus bieti*, *Callithrix jacchus*, etc.

Table: Target species with acession number abd percentage of idenityt with HBB

S.No	Species Name	Accession Number	% Identity with Human HBB
1	Gorilla gorilla	XP_018891709.1	~99%
2	Trachypithecus francoisi	XP_033062959.1	~94%
3	Rhinopithecus bieti	XP_017717110.1	~93%
4	Chlorocebus sabaeus	NP_001316847.1	~96%
5	Callithrix jacchus	XP_002754937.1	~87%
6	Macaca mulatta	NP_001157900.1	~95%
7	Erythrocebus patas	AYO89363.1	~93%

8	Lophocebus albigena	AYO89361.1	~94%
9	Sapajus apella	XP_032096415.1	~90%
10	Aotus nancymaae	XP_012312232.1	~89%
11	Rhinopithecus roxellana	XP_010361646.1	~95%

Data Format:

All sequences were downloaded in **FASTA format** and cleaned for alignment using MEGA X and other downstream tools

Task 2: Pairwise Sequence Alignment

Tools Used: EMBOSS Needle

Objective:

To compare the HBB protein sequence of **Homo sapiens** with:

1. A **closely related species** (*Gorilla gorilla*)
2. A **distantly related species** (*Callithrix jacchus*), to evaluate the degree of conservation.

Tool Used:

EMBOSS Needle

- Algorithm: Needleman-Wunsch (global alignment)
- Substitution matrix: EBLOSUM62
- Gap opening penalty: 10.0
- Gap extension penalty: 0.5

Alignment Results

1. *Homo sapiens* vs. *Gorilla gorilla*

Metric	Value
Aligned Sequences	2
Sequence 1	NP_000509.1 (Human)

Metric	Value
Sequence 2	XP_018891709.1 (Gorilla)
Length	147
Identity	146/147 (99.3%)
Similarity	147/147 (100.0%)
Gaps	0/147 (0.0%)
Score	777.0

NP_000509.1	1	MVHLTPEEKSAVTALWGKVNDEVGGEALGRLLVVYPWTQRFFESFGDLS	50
XP_018891709.	1	MVHLTPEEKSAVTALWGKVNDEVGGEALGRLLVVYPWTQRFFESFGDLS	50
NP_000509.1	51	TPDAVMGNPKVKAHGKKVLGAFSDGLAHLNKGTFATLSELHCDKLHVD	100
XP_018891709.	51	TPDAVMGNPKVKAHGKKVLGAFSDGLAHLNKGTFATLSELHCDKLHVD	100
NP_000509.1	101	PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	147
		:	
XP_018891709.	101	PENFKLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	147

Figure1:Pairwise Sequence Alignment of Human & Gorilla usingEMBOSS Needle .

NP_000509.1	1	MVHLTPEEKSAVTALWGKVNDEVGGEALGRLLVVYPWTQRFFESFGDLS	50
		.	
XP_002754937.	1	MVHLTGEKSAVTALWGKVNDEVGGEALGRLLVVYPWTQRFFESFGDLS	50
NP_000509.1	51	TPDAVMGNPKVKAHGKKVLGAFSDGLAHLNKGTFATLSELHCDKLHVD	100
		. . .	
XP_002754937.	51	TPDAVMNPNPKVKAHGKKVLGAFSDGLTHLDNLKGTF AHLSELHCDKLHVD	100
NP_000509.1	101	PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	147
XP_002754937.	101	PENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH	147

Figure2:Pairwise Sequence Alignment of Human & Callithrix jacchus usingEMBOSS Needle .

Interpretation:

The alignment shows **nearly perfect conservation** of the HBB protein between humans and gorillas, reflecting their **recent divergence** and the strong **evolutionary pressure** to maintain this gene's function in oxygen transport.

2. *Homo sapiens* vs. *Callithrix jacchus*

Metric	Value
Aligned Sequences	2
Sequence 1	NP_000509.1 (Human)
Sequence 2	XP_002754937.1 (Callithrix)
Length	147
Identity	142/147 (96.6%)
Similarity	142/147 (96.6%)
Gaps	0/147 (0.0%)
Score	745.0

Interpretation:

*Although *Callithrix jacchus* is a New World monkey and diverged earlier from humans, the HBB protein still shows **very high conservation**, underscoring the gene's critical role and **functional constraint** across primates.*

Result:

Pairwise alignment confirms that the HBB gene is **highly conserved** across primates. The greater similarity with gorilla reflects closer evolutionary distance, while the slightly lower identity with *Callithrix jacchus* shows how conservation decreases with evolutionary divergence — yet still remains high due to the essential nature of hemoglobin.

Task 3: Multiple Sequence Alignment (MSA)

Tool Used: Clustal Omega (or MUSCLE in MEGA X)

Jalview for visualization

- All 12 sequences aligned

Input:

- A total of **12 HBB protein sequences** from various primate species, including:
 - *Homo sapiens*
 - *Gorilla gorilla*
 - *Macaca mulatta*
 - *Rhinopithecus bieti*
 - *Callithrix jacchus*
 - *Aotus nancymae*
 - *Chlorocebus sabaeus*
 - *Trachypithecus francoisi*
 - *Rhinopithecus roxellana*
 - *Erythrocebus patas*
 - *Sapajus apella*
 - *Lophocebus albigena*

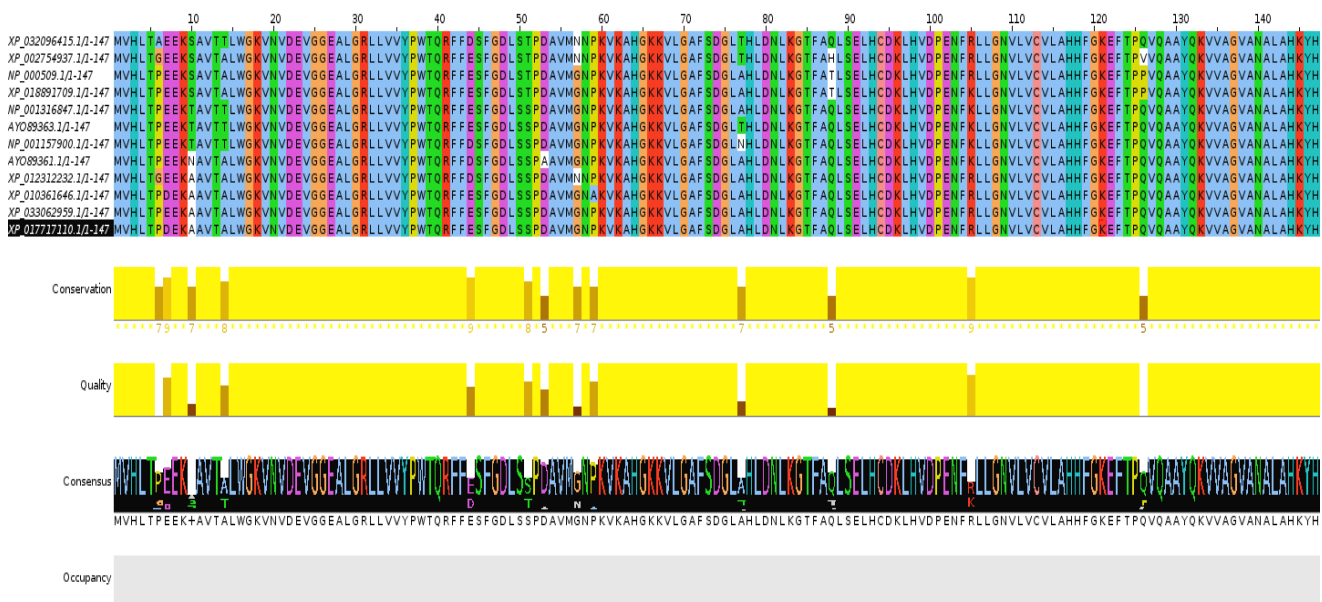


Figure 3: Jalview visualization of All 12 sequences aligned. Amino acid sequences colored by conservation and properties.

Top panel: Amino acid sequences colored by conservation and properties.

- **Conservation plot (yellow bars):** Indicates how well each position is conserved across species.
- **Consensus logo (bottom):** Shows the most conserved residues and their relative frequency.

Observations:

- Residues **Glycine (G)**, **Valine (V)**, **Alanine (A)**, and **Histidine (H)** are consistently conserved at key functional sites.
- The **C-terminal region (positions 128–147)** is **100% conserved** across all species, which is critical for **hemoglobin stability and interaction** with other subunits.

Interpretation:

- These conserved stretches suggest **strong purifying selection** on the HBB gene.
- Functionally essential residues, especially those involved in **heme coordination** and **oxygen transport**, are preserved even in distantly related species like *Callithrix jacchus* and *Aotus nancymae*.

Clustal Omega Alignment Output:

Conserved Regions (with Position Ranges):

Residue Range	Consensus Motif	Comments
1–18	MVHLTPEEKSAVTALWG	Highly conserved N-terminal helix
30–60	VDEVGGEALGRLLVVYPWTQRFFESFGDL	Functionally critical region for heme binding
97–120	CDKLHVDPENFRLLGNVLVCVLAHHFG	Highly conserved interior region
121–147	KEFTPPVQAAYQKVVAGVANALAHKYH	C-terminal tail — completely conserved

Interpretation:

- The alignment reveals **very high conservation**, with several stretches showing full identity across all 12 sequences.

- The final segment (positions **121–147**) shows **complete conservation**, indicating a **strong functional constraint**—these residues likely play essential roles in **stabilizing hemoglobin structure** and **subunit interaction**.
- Minor variations are found in New World monkeys, but do not disrupt the overall structure, indicating evolutionary preservation of function.

Figure4: Clustal Omega alignment of HBB protein sequences across 12 species showing conserved residues (*), strong conservation (:), and moderate conservation (.)

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1 CLUSTAL O(1.2.4) multiple sequence alignment
2
3
4 XP_032096415.1      MVHLTAEKSAVTTTLWGKVNVDDEVGGEALGRLLVVYPWTQRFFDSFGDLSTPDAMNNPK      60
5 XP_002754937.1      MVHLTGEEKSAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMNNPK      60
6 NP_000509.1         MVHLTPEEKSAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPK      60
7 XP_018891709.1      MVHLTPEEKSAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPK      60
8 NP_001316847.1      MVHLTPEEKTAVTTLWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSSPDAMGNPK      60
9 AY089363.1         MVHLTPEEKTAVTTLWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSSPDAMGNPK      60
10 NP_001157900.1      MVHLTPEEKTAVTTLWGKVNVDDEVGGEALGRLLVVYPWTQRFFDSFGDLSSPDAMGNPK      60
11 AY089361.1         MVHLTPEEKNAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFDSFGDLSSPAVMGNPK      60
12 XP_012312232.1      MVHLTGEEKAAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFDSFGDLSSPDAMNNPK      60
13 XP_010361646.1      MVHLPDEKAAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSSPDAMGNAK      60
14 XP_033062959.1      MVHLTPEEKAAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSSPDAMGNPK      60
15 XP_017717110.1      MVHLPDEKAAVTALWGKVNVDDEVGGEALGRLLVVYPWTQRFFESFGDLSSPDAMGNPK      60
16 ***** : ** ***:*****:*****:* ***.* *
17
18 XP_032096415.1      VKAHGKKVLGAFSDGLTHLDNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
19 XP_002754937.1      VKAHGKKVLGAFSDGLTHLDNLKGTFAHLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
20 NP_000509.1         VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
21 XP_018891709.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG      120
22 NP_001316847.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG      120
23 AY089363.1         VKAHGKKVLGAFSDGLTHLDNLKGTFAQLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG      120
24 NP_001157900.1      VKAHGKKVLGAFSDGLNHLNLKGTFAQLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG      120
25 AY089361.1         VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG      120
26 XP_012312232.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
27 XP_010361646.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
28 XP_033062959.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
29 XP_017717110.1      VKAHGKKVLGAFSDGLAHLNLKGTFAQLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG      120
30 *****
31
32 XP_032096415.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
33 XP_002754937.1      KEFTPVVQAAYQKVAGVANALAHKYH 147
34 NP_000509.1         KEFTPQVQAAYQKVAGVANALAHKYH 147
35 XP_018891709.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
36 NP_001316847.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
37 AY089363.1         KEFTPQVQAAYQKVAGVANALAHKYH 147
38 NP_001157900.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
39 AY089361.1         KEFTPQVQAAYQKVAGVANALAHKYH 147
40 XP_012312232.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
41 XP_010361646.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
42 XP_033062959.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
43 XP_017717110.1      KEFTPQVQAAYQKVAGVANALAHKYH 147
44 *****

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Task 4: Sequence Logo Generation

Tool Used:

Skyline/Weblogo – a web-based sequence logo generator.

Sequence Logo Image:



Figure A.



Figure B.



Figure C

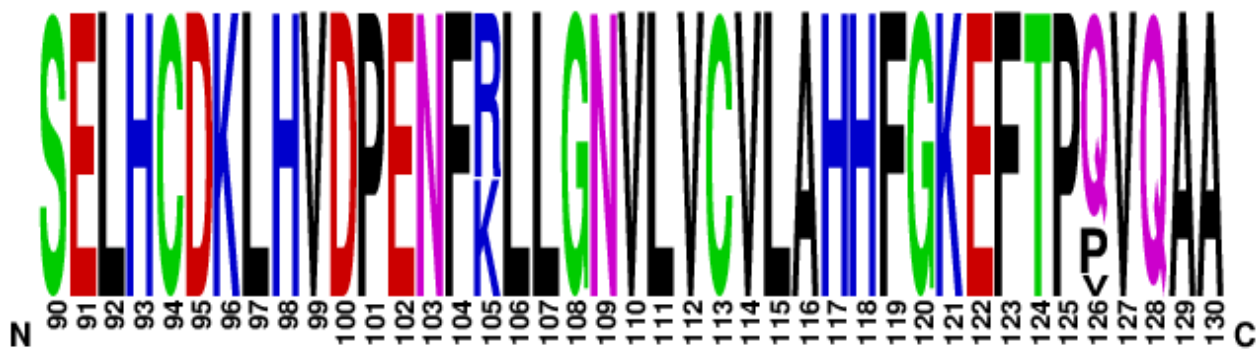


Figure D.

Observations:

- **Tall letters** M, V, H, T, K, Y, in **Figure A** represent **highly conserved residues**, indicating minimal variability at those positions. In **Figure B** tall letters QRSFGDLAVMNK are highly conserved. Similarly in **Figure C** and **D** tall letters GAFSGLNK are conserved.
- The **C-terminal region** (positions ~70–80) shows strong conservation — likely involved in **hemoglobin stability or subunit interaction**.
- Residues such as **G, V, A, L, H** also dominate in mid-sequence, hinting at **core structural roles**.
- **Highly conserved residues are often essential for the structure and function of the HBB protein.**
- Conservation reflects **evolutionary pressure** to maintain amino acids critical for **oxygen binding, folding, and tetramer assembly**.

Task 5: Phylogenetic Tree Construction

Tool Used:

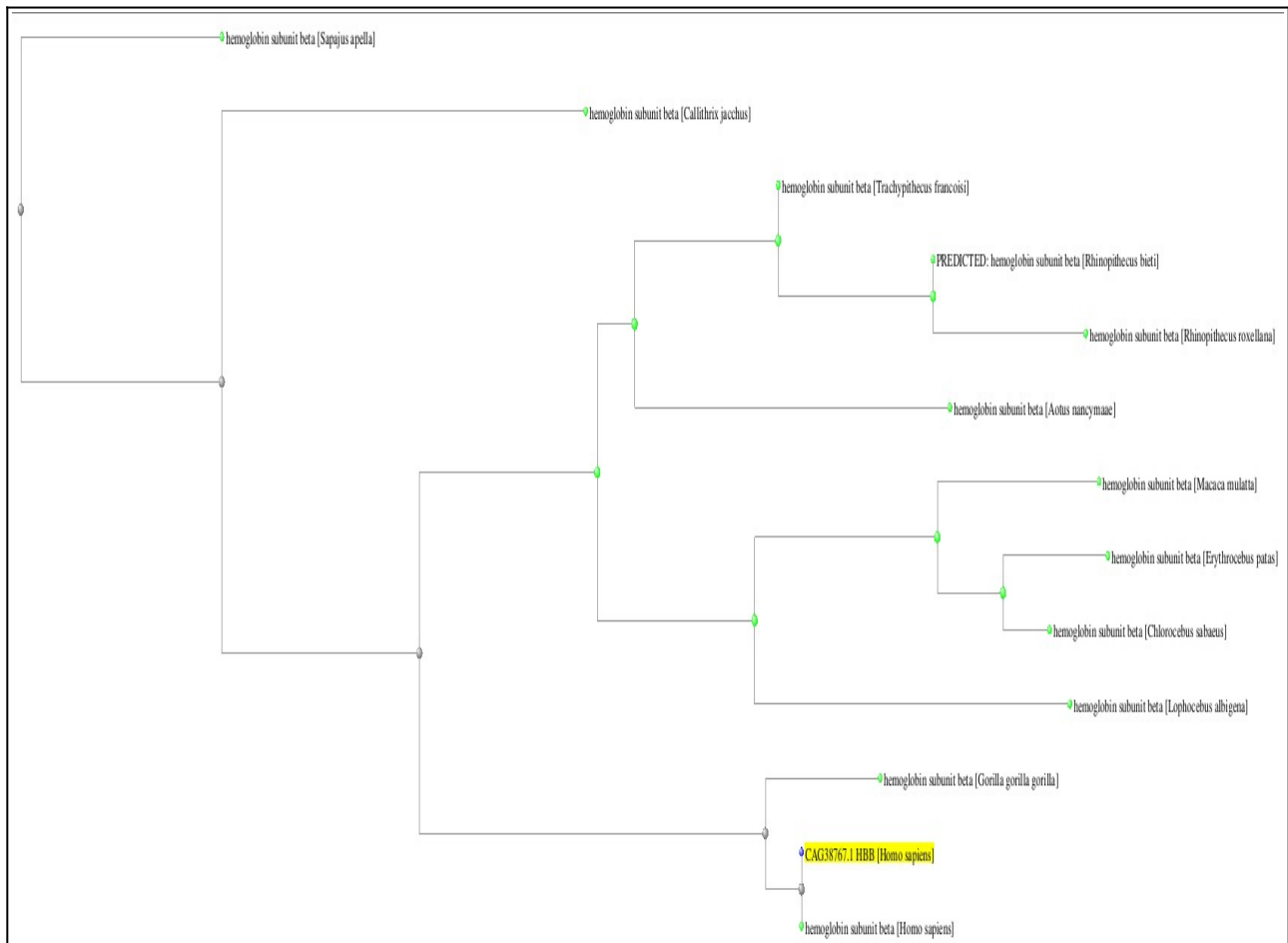
- **MEGA X**
- Method: **Neighbor-Joining (NJ) Tree**

- Bootstrap: **100 replicates**

Input:

- Aligned HBB protein sequences from 12 species (from Clustal Omega MSA)

Figure 5:Phylogenetic Tree



Observations from the Tree:

- **Homo sapiens** clusters **closest to Gorilla gorilla**, confirming their shared recent ancestry.
- *Macaca mulatta*, *Chlorocebus sabaues*, and *Erythrocebus patas* group together, representing **Old World monkeys**.
- *Callithrix jacchus* and *Sapajus apella* form a **distant early branch**, consistent with **New World monkey divergence**.

- *Rhinopithecus* and *Trachypithecus* form a mid-branch clade among **colobine monkeys**.
- All species share a **common ancestor**, confirming that the **HBB gene is conserved** across primates.

Interpretation:

- The tree structure **accurately reflects known primate evolution**:
 - Great apes and humans cluster tightly
 - Old World and New World monkeys are clearly separated
- This supports the hypothesis that the **HBB gene evolved under strong selective pressure**, preserving key functional residues across millions of years.

Conclusion

This mini project successfully explored the evolutionary conservation of the Hemoglobin Beta (HBB) gene across twelve primate species using fundamental bioinformatics tools. Starting with sequence retrieval and BLAST analysis, we identified highly similar HBB protein sequences with high identity scores, particularly between humans and closely related primates like gorillas. Pairwise alignments confirmed this conservation, showing near-perfect identity with gorilla and high similarity even in more distantly related species like *Callithrix jacchus*. Multiple sequence alignment revealed strongly conserved motifs across the entire protein, which were further visualized through a sequence logo generated using Skylign OR Weblogo—highlighting key functional residues maintained across species. Finally, phylogenetic analysis using MEGA X produced a tree that aligned with known evolutionary relationships, clearly distinguishing great apes, Old World monkeys, and New World monkeys. Together, these analyses demonstrate that the HBB gene is highly conserved due to its essential physiological role, making it a reliable molecular marker for studying evolutionary relationships among primates.